CSC2002S

ToC Assignment 2025

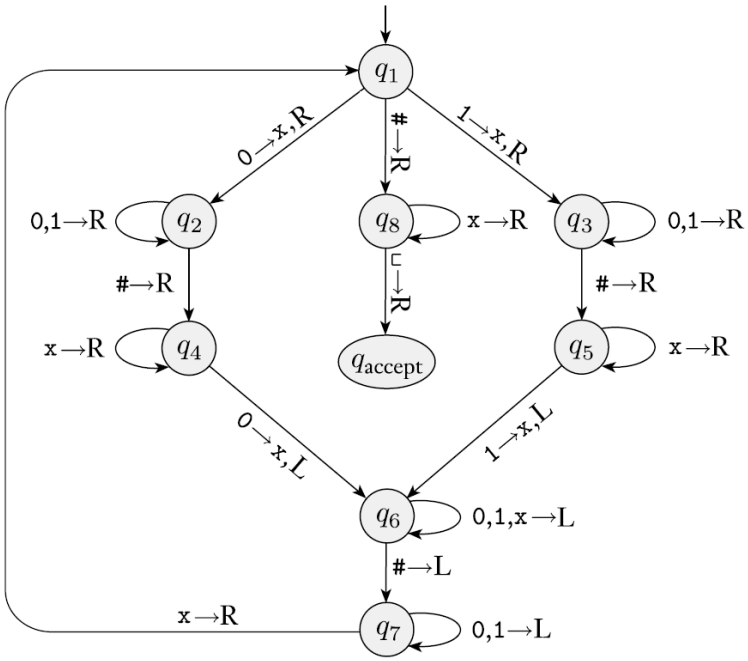
1) Design a basic – i.e., one tape, one track, Turing Machine that, given some sequence of a's, b's and c's as input string on its tape:
   i) accepts all words of the form abac$^+$aba, (note: c$^+$ means one or more occurrences of c), and
   ii) 2. loops forever on all words that start with b. (5)

2) Create a multi-tape Turing machine that accepts Strings of the form, xYreverse(x)!, where x contains a pattern of 1s and 0s. e.g. 0101Y1010! is a valid string. (5)

Hint:

- use a two tape machine
- the second Tape will play the role of a stack.
- both tapes should be empty (all blanks) after processing through a valid input
- An optimal solution has only 3 states.

3) Consider the following TM M1:

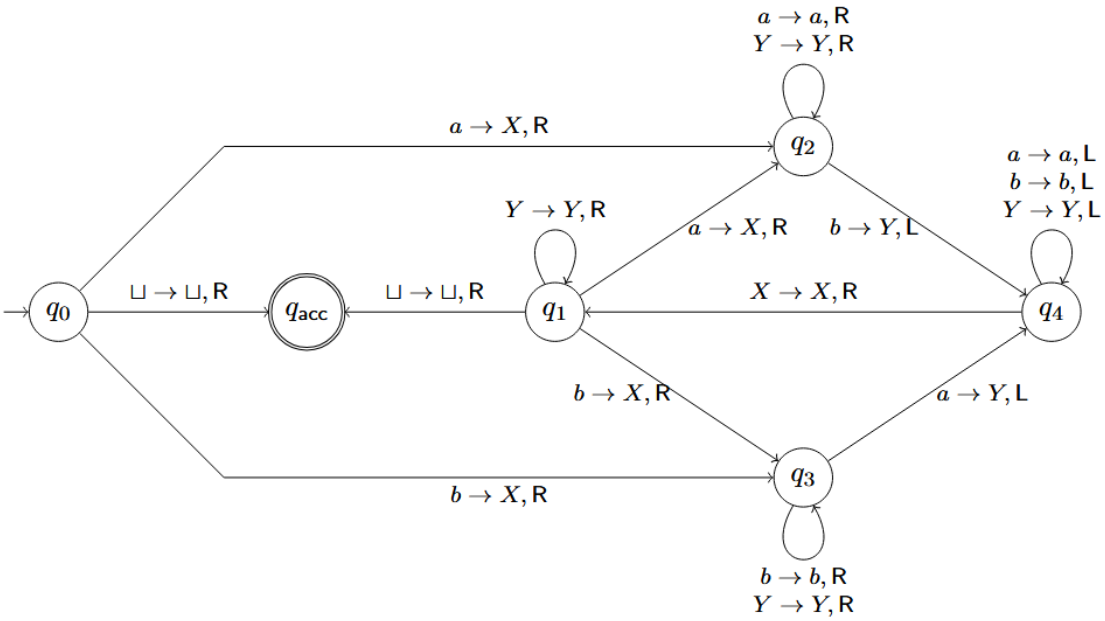


Note: the label 0⟶x, R appears on the transition from q1 to q2. This label signifies that when in state q1 with the head reading 0, the machine goes to state q2, writes x, and moves the head to the right. You will also find the label 0,1⟶R on the transition going from q3 to itself. That label means that the machine stays in q3 and moves to the right when it reads a 0 or a 1 in state q3. It doesn't change the symbol on the tape.

Give the sequence of configurations (instantaneous descriptions) that M1 enters when started on the following input strings.

i) 01#01 (3)
ii) 0#0# (2)

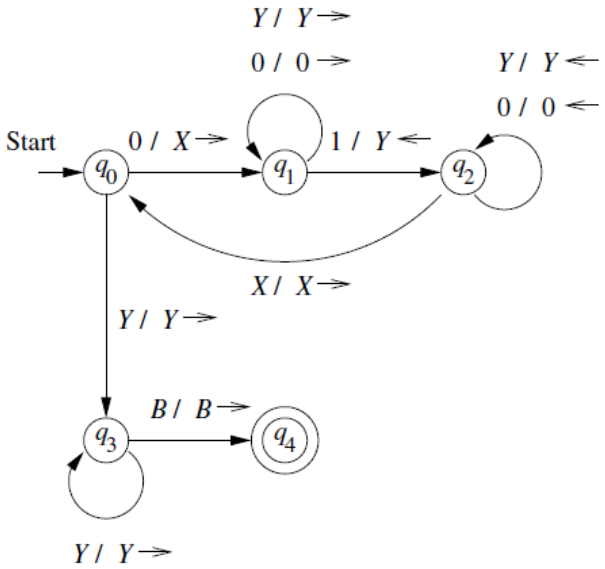
4) Consider the following Turing Machine M with input alphabet Σ = {a, b}.



i) Give the formal definition of δ (the transition function). Show all transitions. (4)

ii) Describe the language recognized by M. (1)

5) Consider the following TM:



In the table below, give the codes for each of the rules above, using the string: $0^i 10^j 10^k 10^l 10^m$, where i,k = {1(for q0),2 (for q1),3,4,5}; j,l = {1 (for input 0), 2 (for 1), 3 (for X), 4 (for Y), 5 (for B)}; and m = {1 (for L), 2 (for R)} (5)

State	Input	Code
q0	0	
q0	Y	
q1	0	
q1	1	
q1	Y	
q2	0	
q2	X	
q2	Y	
q3	Y	
q3	B	